

ORIGINAL ARTICLE

Minimal risk of tilapia lake virus transmission via frozen tilapia fillets

Worrayanee Thammatorn¹ | Pattarasuda Rawiwan^{1,2}  | Win Surachetpong^{1,2} ¹Department of Veterinary Microbiology and Immunology, Faculty of Veterinary Medicine, Kasetsart University, Bangkok, Thailand²Center for Advanced Studies for Agriculture and Food, Institute for Advanced Studies, Kasetsart University, Bangkok, Thailand**Correspondence**Win Surachetpong, Department of Veterinary Microbiology and Immunology, Faculty of Veterinary Medicine, Kasetsart University, Bangkok, Thailand.
Email: fvetwsp@ku.ac.th**Funding information**

The Center for Advanced Studies for Agriculture and Food, Institute for Advanced Studies, Kasetsart University; The Higher Education Research Promotion and National Research University Project of Thailand; Office of the Higher Education Commission, Ministry of Education, Thailand; Faculty of Veterinary Medicine, Kasetsart University, Bangkok, Thailand

Abstract

Recent outbreaks of a novel tilapia lake virus (TiLV) have raised concerns regarding the international spread of TiLV in frozen tilapia products. This study investigated the potential risks of frozen tilapia fillet as a source of TiLV transmission. It revealed that TiLV genomic RNA could be detected in tilapia fillet and the virus isolated from non-frozen and frozen fillets with clinical TiLV infection stored up to 28 days caused a cytopathic effect (CPE) formation in the susceptible cell line in vitro. However, frozen fillets from clinical TiLV infection stored for 90 and 120 days did not cause CPE in the susceptible cell line. Similarly, CPE was not observed in TiLV isolated from subclinically TiLV-infected fish fillets. In addition, in vivo bioassay revealed that despite the presence of TiLV isolated from subclinically TiLV-infected fillet stored at -20°C for 14 days, there was no evidence of TiLV disease in naïve red hybrid tilapia based on the absence of clinical signs and mortality and without the detection of TiLV genomic RNA using reverse transcription-quantitative polymerase chain reaction assay. Collectively, these findings suggested minimal risk of transmission of TiLV via frozen tilapia fillets.

KEYWORDS

frozen fillet, persistence, risk, tilapia, tilapia lake virus

1 | INTRODUCTION

Tilapia lake virus (TiLV) was first identified in wild and farm-raised tilapia showing clinical signs of skin discoloration, ocular alterations and lethargy in Israel (Eyngor et al., 2014). Subsequently, TiLV was found in many countries including Ecuador (Bacharach et al., 2016), Colombia (Kembou Tsofack et al., 2017), Egypt (Fathi et al., 2017; Nicholson et al., 2017), Thailand (Dong et al., 2017; Surachetpong et al., 2017), India (Behera et al., 2018), Indonesia (Koesharyani, Gardenia, Widowati, Khumaira, & Rustianti, 2018), Uganda and Tanzania (Mugimba et al., 2018), and Malaysia (Amal et al., 2018). In clinically infected fish, TiLV has been detected in several organs including liver, spleen, anterior kidney, gills, heart as well as muscle (Dong et al., 2017; Tattiyapong, Dachavichitlead, & Surachetpong, 2017; Tattiyapong, Sirikanchana, & Surachetpong, 2018). To date, various methods have been used for the diagnosis of TiLV such as reverse transcription-polymerase chain reaction (RT-PCR), RT-quantitative PCR (RT-qPCR), histopathology and viral isolation in

susceptible cells (Behera et al., 2018; Jaemwimol et al., 2018; Kembou Tsofack et al., 2017; Tattiyapong et al., 2018; Waiyamitra et al., 2018).

In 2015, more than 6.4 million tonnes of tilapia aquaculture were produced globally and were valued at USD 1.8 billion for trading (FAO, 2017a). Although there have been many infectious diseases that affect tilapia production, there has been little evidence on the impact on these diseases to tilapia trade. However, recent outbreaks of TiLV have raised concerns on whether tilapia fillet and frozen whole fish could be a potential source of viral transmission (FAO, 2017a). Thus, it is important to investigate the potential risk of virus spread via international trade of frozen tilapia fillets. In addition, the effects of freezing temperature and time on TiLV persistence and risk of contamination to the importing countries have not been extensively investigated.

The objective of this study was to determine the effects of commercial freezing temperature (-20°C) and storage times on the persistence and infectivity of TiLV in frozen fish fillets. The TiLV persistence and viability in frozen fish fillets were analysed using

RT-qPCR, viral isolation in cell culture and bioassay in susceptible red hybrid tilapia.

2 | MATERIALS AND METHODS

2.1 | Animal and ethical statement

Red hybrid tilapia (*Oreochromis* spp.) were obtained from a local fish farm in Saraburi province, Thailand, with a history of good farm management. Fish were allowed to acclimatize at 28°C in a 400-L aquarium tank with dechlorinated water for 1 week at the Animal Research Facility, Faculty of Veterinary Medicine, Kasetsart University, Bangkok, Thailand. During acclimatization, the water was changed daily and the temperature ($28 \pm 0.29^\circ\text{C}$), dissolved oxygen (DO; 5.92 ± 0.67 ppm), pH (7.40 ± 0.17) and total ammonia (0.25 ± 0.05 ppm) were also checked and recorded daily. All laboratory protocols and animals used were approved by the Kasetsart Institutional Animal Care and Use Committee under permission number ACKU61-VET-009.

2.1.1 | Experimental design and sample collection criteria

In a preliminary experiment, naïve red hybrid tilapia ($n = 10$; weight = 10 ± 0.5 g) were intraperitoneally (IP) injected with 50 μl of TiLV strain VET-KUTV01, isolated from red hybrid tilapia obtained from Pathum Thani province, Thailand (Tattiyapong et al., 2017), in a titre of 1×10^5 tissue culture infective dose by 50% (TCID₅₀)/ml. At 6 days post-infection (dpi), 7 clinically TiLV-infected fish were killed with an overdose of clove oil (Aquanes; Better Pharma, Bangkok, Thailand) and 3 separate fillets (100 mg each) from each fish were collected and stored at commercial freezing temperature (-20°C) for 0, 14 and 28 days. The TiLV genomic RNA and TiLV infectivity were determined using RT-qPCR assay and virus isolation, respectively.

In a further experiment, naïve red hybrid tilapia ($n = 10$; weight = 10 ± 0.5 g) were IP injected with 50 μl of TiLV strain VET-KUTV01 in a titre of 1×10^5 TCID₅₀/ml. At 6 dpi, 5 clinically TiLV-infected fish were killed with an overdose of clove oil and 2 separate fillets (100 mg each) from each fish were collected and stored at -20°C for 90 and 120 days. At 90 and 120 days post-freezing, the TiLV loads in the fillets were determined using RT-qPCR assay and virus isolation.

In a separate experiment, naïve red hybrid tilapia ($n = 10$; weight = 10 ± 0.5 g) were IP injected with 50 μl of TiLV strain VET-KUTV01 in a titre of 1×10^5 TCID₅₀/ml. At 6 dpi, 6 subclinically TiLV-infected fish were killed with an overdose of clove oil and 3 separate fillets (100 mg each) from each fish were collected and stored at -20°C for 0, 14 and 28 days. At 0, 14 and 28 days post-freezing, the TiLV loads in the fillets were determined using RT-qPCR assay and virus isolation.

2.2 | RNA extraction and cDNA synthesis

In brief, RNA was isolated from 100 mg of the trimmed red hybrid tilapia fillet using 1 ml of commercial TRIzol® reagent (Thermo

Fisher Science Corp.; Carlsbad, CA, USA) in a 1.5-ml microcentrifuge tube. The two separated phases were homogenized using a pestle homogenizer until homogeneous. Thereafter, 200 μl of chloroform (EMSURE®; Merck KGaA; Darmstadt, Germany) was added into the tube and gently mixed followed by incubation at room temperature for 3 min. The solution was then centrifuged using a microcentrifuge (5145R; Eppendorf GmbH; Hamburg, Germany) at 4°C at $12,000 \times g$ for 15 min. Afterwards, 500 μl of the upper aqueous phase was pipetted into a new tube. Next, 200 μl of isopropanol was added and incubated at -20°C for 30 min before centrifugation at 4°C $12,000 \times g$ for 15 min. After discarding the supernatant, the white RNA pellet was washed with 75% ethanol by centrifugation at 4°C $10,000 \times g$ for 15 min. The supernatant was discarded and the residual ethanol was disposed using a pipette before the RNA pellet was allowed to air-dry at room temperature for 5 min. The semi-dried RNA pellet was resuspended in 60 μl DNase-free water at $55\text{--}60^\circ\text{C}$. Lastly, the concentration and quality of the total RNA were quantified using a spectrophotometer (NanoDrop2000; Thermo Fisher Scientific Corp.) and diluted the concentration to 200 ng/ μl using DNase-free water.

A ReverTraAce® qPCR RT Master Mix kit (Toyobo, Osaka, Japan) was used for cDNA transcription. Briefly, 1 μg of RNA was vigorously mixed with 5x RT Master Mix and the reaction was made up to 10 μl with nuclease-free water using a vortex (Genie® II; Scientific Industries Inc.; Bohemia, NY, USA) and spun-down using a microcentrifuge (Corning™ LSE mini microcentrifuge; Thermo Fisher Scientific Corp.; Carlsbad, CA, USA). Subsequently, the reaction was incubated at 65°C for 5 min followed by 42°C for 60 min and 98°C for 5 min in a thermal cycler (T100; Bio-Rad Laboratories Inc.; Hercules, CA, USA). The cDNA was then used immediately or kept at -20°C for further analysis.

2.3 | RT-qPCR assay

The viral load from fish fillet was determined using SYBR green-based RT-qPCR assay (Tattiyapong et al., 2018). In brief, 200 ng of cDNA in a 10 μl reaction was performed consisting of 5 μl of 2x iTaq™ universal SYBR Supermix (Bio-Rad Laboratories Inc.), 0.3 μl of forward primer (TiLV-112F: 5'-CTGAGCTAAAGAGGCAATATGGATT-3'), and 0.3 μl of reverse primer (TiLV-112R: 5'-CGTGCCTACTCGTTTCAGTATAAGTTCT-3'). The reaction was carried out in a real-time PCR thermocycler (CFX96 Touch™; Bio-Rad Laboratories Inc.). The conditions used were as follows: 1 cycle of 95°C for 3 min, 40 cycles of 95°C for 10 s and 60°C for 30 s. At the end of the reaction, melting curve analysis was performed at $65\text{--}95^\circ\text{C}$ with an increment of 0.5°C per 5 s according to Tattiyapong et al. (2018).

2.4 | Viral isolation in cell culture

The red hybrid tilapia fillet sample was trimmed and added into a microcentrifuge tube after which 900 μl of Hank's balanced salt solution (Corning® HBSS; Mediatech Inc.; Manassas, VA, USA) was added into the tube. The fillet was homogenized and centrifuged

at $3,000 \times g$ at 4°C for 10 min. Then, the supernatant was filtered through a $0.22\text{-}\mu\text{m}$ syringe filter (Corning Inc.; Corning, NY, USA) and the filtered virus was kept at -80°C for further analysis. The filtered virus was inoculated in E-11 cells, which were purchased from the European Collection of Authenticated Cell Cultures (ECACC; Porton Down, UK). E-11 cells were maintained in Leibovitz L-15 (Merck KGaA; Darmstadt, Germany) supplemented with 5% inactivated foetal bovine serum (Thermo Fisher Science Corp.; Carlsbad, CA, USA) and cultured at 25°C in a 96-well plate for daily observation on the development of the cytopathic effect (CPE) for 14 days.

2.5 | In vivo bioassay

In the first experiment, naïve red hybrid tilapia ($n = 10$; weight = 20 ± 0.5 g) were IP injected with $50\text{ }\mu\text{l}$ of TiLV strain VET-KUTV01 in a titre of 1×10^4 TCID₅₀/ml. All fish were killed at 6 dpi using an overdose of clove oil. The virus was isolated from 100 mg of fillet according to section 2.5 above, and the virus was quantified using RT-qPCR assay according to section 2.4 above. The level of TiLV in each fish fillet sample was classified as: (a) high infection with clinically TiLV-infected fish with a threshold cycle (C_t) value < 26 and (b) low infection with subclinically TiLV-infected fish with a $C_t \geq 26$. Thereafter, a group of naïve red hybrid tilapia ($n = 16$; weight = 10 ± 0.5 g) were IP injected with $50\text{ }\mu\text{l}$ of the aforementioned isolated virus (mean C_t value = 20.89) which was referred to as high infection. In addition, another group ($n = 16$; weight = 10 ± 0.5 g) of fish were IP injected with $50\text{ }\mu\text{l}$ of the aforementioned isolated viruses (mean C_t value of 32.40) which was referred to as low infection. The clinical signs of TiLV infection and mortality rate were monitored daily, and fish ($n = 3$) were randomly collected for the determination of C_t values from the fillets using RT-qPCR assay at 6 and 14 dpi.

In the second experiment, isolated virus was prepared from fish fillets of subclinical fish with C_t values of 30.30 (Rep. 1), 31.46 (Rep. 2) and 32.65 (Rep. 3) killed at 6 dpi and stored at -20°C for 14 days.

Naïve red hybrid tilapia ($n = 48$; weight = 10 ± 0.5 g) were separated into 3 equal groups, and each group was IP injected with isolated virus from Rep. 1, Rep. 2 and Rep. 3, respectively. The clinical signs of TiLV infection and mortality rate were recorded daily. At 6 and 14 dpi, fish ($n = 3$) were randomly collected for the quantification of C_t values from the fillet and liver using RT-qPCR assay (Supporting Information Figure S1).

2.6 | Statistical analysis

Differences in the levels of TiLV genomic RNA based on RT-qPCR were tested using the multiple comparison post hoc test and one-way analysis of variance. All means were tested for least significant differences at the 95% confidence interval. Where there were 2 mean groups of analyses, the independent sample *t* test was used to calculate significant difference at the 95% confidence interval. Analysis was performed using the SPSS[®] statistical software (version 22.0; IBM[®] Corp.; Armonk, NY, USA). Any *p* value < 0.05 was considered significantly different ($p < 0.05$).

3 | RESULTS

3.1 | In vitro analysis

Tilapia lake virus genomic RNA was detected in non-frozen tilapia fillets in all clinically TiLV-infected fish at 0 dpi. Similarly, TiLV genomic RNA persisted in the fillets frozen at -20°C at 14 and 28 dpi, with C_t values ranging between 17.22 and 24.68, and 16.22 and 24.48, respectively (Table 1). However, there were no statistical differences amongst the mean C_t values of TiLV at 0, 14 and 28 dpi. Further analysis of non-frozen fillets indicated that the virus was viable, resulting in CPE formation in E-11 cells within 3–6 days post-inoculation. In contrast, the virus in fillets stored at -20°C for 14 and 28 days delayed CPE formation with a longer period (5–10 days) as shown in Table 1. Notably, one of the frozen fillets

TABLE 1 Detection of TiLV genomic RNA (threshold cycle value; C_t) in frozen tilapia fillet from clinically infected tilapia stored at -20°C for 0, 14 and 28 days

Fish no.	C_t values			CPE formation (days post-inoculation)		
	Day 0	Day 14	Day 28	Day 0	Day 14	Day 28
1	17.04	17.22	16.22	3	10	9
2	17.73	17.79	16.39	3	5	9
3	17.36	17.31	17.41	3	10	9
4	17.45	18.68	16.83	5	5	7
5	20.36	24.31	24.48	6	–	–
6	17.89	21.38	16.75	4	6	6
7	22.43	24.68	20.36	6	6	6
Mean $\pm$ SD	$18.61 \pm 2.01^*$	$20.20 \pm 3.26^*$	$18.35 \pm 3.05^*$			

Notes. Each frozen tilapia fillet stored at -20°C for 0, 14 and 28 days was filtered and inoculated with E-11 cells to evaluate CPE formation. CPEs were observed for 14 days post-inoculation.

C_t , threshold cycle; CPE, cytopathic effect; (–), not detected; SD, standard deviation.

Mean $\pm$ SD ($n = 7$).

Level of significance, $*p < 0.05$.

(fish no. 5) stored for 14 and 28 days did not produce CPE. In subsequent study, the TiLV genomic RNA was also detected in fillets from clinically TiLV-infected red hybrid tilapia stored at -20°C for 90 and 120 days (Table 2). The C_t values of TiLV from the fillets stored for 90 and 120 days ranged from 17.99 to 26.80 and from 23.51 to 29.27, respectively. There was no significant difference between the mean C_t values of TiLV between these two storage conditions. Importantly, no CPE was found in E-11 cells inoculated with virus from frozen fillet stored at 90 and 120 days (Table 2).

The results from the series of experiment performed to evaluate the persistence of TiLV in subclinically TiLV-infected fish fillets indicated that TiLV genomic RNA could be detected in both non-frozen and frozen fillets stored for 0, 14 and 28 days with C_t values ranging from 26.49 to 36.25 (Table 3). There was a significant difference between the mean C_t values of TiLV in the frozen fillets stored for 14 and 28 days; however, no significant difference was detected for the mean C_t values of virus in non-frozen fillets and frozen fillets stored for 14 days. Importantly, neither non-frozen fillets nor frozen fillets from subclinically TiLV-infected fish produced any CPE.

3.2 | In vivo bioassay

The virus was isolated from non-frozen and frozen tilapia fillets and IP injected into naïve tilapia. The results revealed that TiLV isolated from non-frozen fillets of clinically and subclinically TiLV-infected fish with C_t values of 20.89 and 32.40, respectively, produced clinical signs of TiLV infection with cumulative mortality rates of 80% and 30%, respectively (Figure 1). The clinical signs of TiLV infection including loss of appetite, skin redness, abdominal swelling and exophthalmos were clearly observed at 6 dpi. The cumulative mortality rate was evident from 6 until 14 dpi in fish injected with TiLV isolated from clinically infected fish fillet, while the onset of mortality of fish injected with TiLV isolated from subclinically infected fish fillet was evident from 7 until 11 dpi (Figure 1). Subsequently, the fillets of three subclinically infected fish stored at -20°C for 14 days with C_t values of 30.30, 31.46 and 32.65, respectively, were injected into the 3 groups of naïve tilapia ($n = 16$). The findings showed that

no TiLV genomic RNA was detected in either the livers or fillets of challenged fish at 6 and 14 dpi (Supporting Information Table S1). No mortality was recorded throughout 21 days of observation in all 3 replicates of fish injected with TiLV isolated from frozen subclinically infected fillet (Figure 1).

4 | DISCUSSION

Tilapia are amongst the important freshwater fish consumed globally with international trade of frozen whole and frozen fillet being approximately 180,000 tonnes between January and March 2017 in which the United States, Mexico, Ivory Coast and Iran being the major importers while China, the Republic of China (Taiwan), Thailand and Indonesia were the leading exporters (FAO, 2017b). However, recent outbreaks of TiLV have raised concern regarding its impact on tilapia production worldwide. Currently, there is no specific antiviral compound or vaccine for the prevention of TiLV infection. It has been suggested that limiting the spread of the virus requires the implementation of strict biosecurity and risk management on the importation of live tilapia and tilapia products. However, it is doubtful whether viable TiLV particles can survive harsh conditions in tilapia commodities such as freezing temperatures (FAO, 2017a). Consequently, it was critical to study the freezing duration due to the sudden incidence of TiLV disease outbreaks worldwide through the international trade.

The current study showed that TiLV isolates from non-frozen fillet of clinically TiLV-infected tilapia caused CPE development in E-11 cells, suggesting that the virus was still viable in fillets of infected fish. However, the viability of TiLV isolates from clinically TiLV-infected tilapia fillet might be retarded by commercial freezing temperature and extended frozen storage as being observed on the later dates of CPE formation at 14 and 28 days post-freezing and no development of CPE at 90 and 120 days post-freezing. Similarly, research on Alaskan marine fish showed that freezing Pacific hake (*Merluccius productus*), Pacific herring (*Clupea pallasii*) and Pollock (*Theragra chalcogramma*) could reduce the infectivity of viral haemorrhagic

Fish no.	C_t values		CPE formation (days post-inoculation)	
	Day 90	Day 120	Day 90	Day 120
1	26.80	29.27	–	–
2	17.99	24.13	–	–
3	24.72	23.51	–	–
4	20.27	N/A	–	N/A
5	18.21	N/A	–	N/A
Mean $\pm$ SD	21.60 $\pm$ 3.97*	25.64 $\pm$ 3.16*		

TABLE 2 Detection of TiLV genomic RNA (threshold cycle value; C_t) in frozen tilapia fillet from clinically infected tilapia stored at -20°C for 90 and 120 days

Notes. Each frozen tilapia fillet stored at -20°C for 90 and 120 days was filtered and inoculated with E-11 cells to evaluate CPE formation. CPE was observed for 14 days post-inoculation.

C_t , threshold cycle; CPE, cytopathic effect; (–), not detected; SD, standard deviation;

Mean $\pm$ SD ($n = 5$).

Level of significance, * $p < 0.05$.

TABLE 3 Detection of TiLV genomic RNA (threshold cycle value; C_t) in frozen tilapia fillet from subclinically infected tilapia stored at -20°C for 0, 14 and 28 days

Fish no.	C_t values			CPE formation (days post-inoculation)		
	Day 0	Day 14	Day 28	Day 0	Day 14	Day 28
1	31.09	29.64	26.49	–	–	–
2	27.31	35.40	26.50	–	–	–
3	34.49	35.08	31.98	–	–	–
4	28.85	31.99	28.36	–	–	–
5	32.03	36.25	26.72	–	–	–
6	27.15	30.80	27.37	–	–	–
Mean $\pm$ SD	30.15 $\pm$ 2.90***	33.19 $\pm$ 2.74*	27.90 $\pm$ 2.12***			

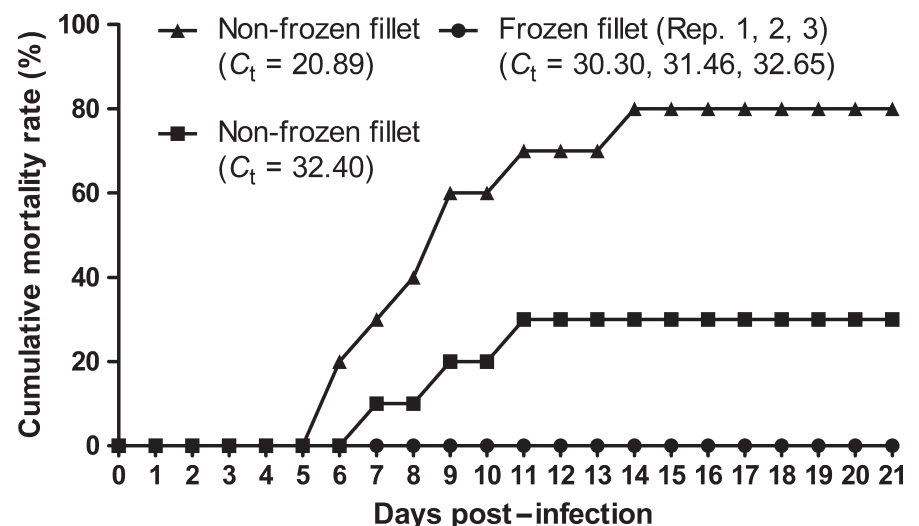
Notes. Each frozen tilapia fillet stored at -20°C for 0, 14 and 28 days was filtered and inoculated with E-11 cells to evaluate CPE formation. CPE was observed for 14 days post-inoculation.

C_t , threshold cycle; CPE, cytopathic effect; (–), not detected; SD, standard deviation;

Mean $\pm$ SD ($n = 6$)

The *, **, and *** indicate the significant difference of mean $\pm$ SD of C_t value of each storage day group; Level of significance, $p < 0.05$.

FIGURE 1 Cumulative mortality rate of naïve red hybrid tilapia ($n = 10$) injected with TiLV isolated from non-frozen fillet from clinically infected tilapia ($\blacktriangle$, threshold cycle (C_t) = 20.89), TiLV isolated from non-frozen fillet from subclinically infected tilapia ($\blacksquare$, $C_t = 32.40$) and TiLV isolated from frozen fillet from subclinically infected tilapia ($\bullet$, $C_t = 30.30, 31.46$ and 32.65 for replicate (Rep.1, Rep.2 and Rep. 3, respectively). The mortality was recorded daily for 21 dpi. The clinically TiLV-infected fish were classified using $C_t < 26$, and the subclinically TiLV-infected fish was classified using $C_t \geq 26$



septicaemia virus (VHSV) by 1,000-folds (Meyers, Short, & Lipson, 1999). Furthermore, if tissues with high virus concentration were treated using freezing, such treatment might help to reduce the amount of virus but might not eliminate it to an undetectable level (Kocan, Hershberger, Elder, & Winton, 2001; Kocan et al., 1997).

More importantly, the TiLV isolated from subclinically TiLV-infected tilapia fillets frozen for 0, 14 and 28 days were not infective since no development of CPE was observed. Likewise, the bioassay study suggested that no TiLV genomic RNA was found in the liver and fillet of tilapia infected with TiLV isolated from subclinically infected fish fillets frozen for 14 days (Supporting Information Table S1). Moreover, the mortality was not detected throughout 21 days of observation. This suggested that the 14 days post-freezing fillets from subclinically infected tilapia were avirulent and free from viable virus particles. Arkush et al. (2006) reported that commercial freezing temperature (-20°C) could reduce VHSV infectivity in tissues of challenged Pacific sardine (*Sardinops sagax*). Additionally, the aquatic virus presented in the tissues may have lost its capability of infection due to ice crystal

formation during the freezing process damaging the viral envelop and nucleotides (Durand, Tang, & Lightner, 2000) which was consistent with Wertheimer and Winton (1982) who reported that infectious haematopoietic necrosis virus (IHNV) may not have jeopardized chinook salmon (*Oncorhynchus tshawytscha*) if stored in the freezer.

Given the concern from countries having a TiLV-negative status and their strict enforcement of regulations on the importation of live tilapia, the risk from importation of whole frozen fish or frozen fillet remains controversial and largely unknown. Our study provides the first evidence to support risk analysis of TiLV introduction into importing countries and will help the relevant authorities decide on allowing the importation of tilapia products. As the risk of TiLV infection in frozen fillet is very low and considering that only healthy fish will be delivered to processing plants, it is unlikely that the infective virus could be present in tilapia fillets. Although the data presented in this study indicated very low risk of the introduction of TiLV through frozen tilapia fillets, it is possible that other contaminated visceral organs (such as the liver,

spleen or anterior kidney) containing a high viral load in eviscerated products may pose a potential risk of spreading the virus. Furthermore, cleaning fish products using appropriate disinfectants or treatment methods will additionally reduce the amount of infective virus and limit the risk of virus contamination (OIE, 2009; Smail, Grant, Simpson, Bain, & Hastings, 2004).

In summary, our study indicated that TiLV isolated from clinically infected tilapia fillet could remain viable up to 28 days at -20°C . However, the fillet collected from subclinically infected tilapia stored at -20°C at least 14 days did not pose any potential risk or contain viable TiLV particles even though genomic RNA could be detected using a highly sensitive molecular method. Although the international trade of fish products presents a potential risk to importing countries through the accidental importation of products contaminated with fish pathogens, the current study revealed that tilapia fillets stored at the commercial freezing temperature have a very low risk of spreading the virus. However, good practices such as exporting products from healthy fish will further reduce the risk of TiLV spreading via the trading of fish products and there should be more emphasis placed on the study. As such, information in this study should be considered by the competent authorities for the risk evaluation regarding their importation of tilapia products.

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CONFLICT OF INTERESTS

The authors declare no conflict of interests.

ORCID

Pattarasuda Rawiwan  <http://orcid.org/0000-0002-0308-1112>

Win Surachetpong  <http://orcid.org/0000-0002-5707-3476>

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SUPPORTING INFORMATION

Additional supporting information may be found online in the Supporting Information section at the end of the article.

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